Decision Tree

A Full Step-by-Step Exercise with Math Calculations

Dataset
Play Tennis

Method
ID3 / Entropy

Goal
Classify Outlook

Step 0 — The Dataset

We want to predict whether to PLAY TENNIS based on 4 weather features:

Outlook	Temperature	Humidity	Wind

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Total: 14 days ✓
Play = 9 (Yes) ✗ No Play = 5 (No)
Target: Predict "Play Tennis"

Step 1 — What is Entropy?

Entropy Formula:

$$H(S) = -\sum p_i \cdot \log_2(p_i) \qquad \text{where } p_i = \text{proportion of class } i \text{ in dataset } S$$

● H = 0 → Pure

All examples belong to ONE class.
No uncertainty at all.

⍰ H = 1 → Max Impure

Classes are perfectly mixed (50/50).
Maximum uncertainty.

● Goal: Minimize H

Decision Tree splits data to
reduce entropy as fast as possible.

Calculating Root Entropy H(S)

Dataset S has 9 Yes and 5 No → Total = 14

$p(\text{Yes}) = 9/14 = 0.643 \qquad p(\text{No}) = 5/14 = 0.357$

$H(S) = -(0.643 \times \log_2 0.643) - (0.357 \times \log_2 0.357) = -(0.643 \times -0.637) - (0.357 \times -1.486)$

$H(S) = 0.410 + 0.531 = 0.940$ bits ← This is our starting impurity!

Step 2 — Information Gain (IG)

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IG(S, A) = H(S)  −  Σ  (|Sv| / |S|) × H(Sv)
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We pick the attribute A with the HIGHEST IG

We will now compute IG for each of the 4 features:

Outlook	Temperature	Humidity	Wind
IG = 0.247	IG = 0.029	IG = 0.152	IG = 0.048
⍰ BEST	⍰ 3rd	⍰ 2nd	4th

How we compute IG for OUTLOOK (next slide):

1. Split data by each value of Outlook: Sunny (5 days), Overcast (4 days), Rain (5 days)
2. Compute entropy H for each subset
3. Compute weighted average: Σ (size_subset / total) × H(subset)
4. IG = H(root) − weighted average = 0.940 − 0.693 = 0.247

Step 3 — IG Calculation: OUTLOOK (Detailed)

☀ Sunny (5 days)

D1: No

D2: No

D8: No

D9: Yes

D11: Yes

Yes=2 No=3 Total=5

$H = -(2/5)\log_2(2/5) - (3/5)\log_2(3/5)$

$= -(0.4\times-1.322) - (0.6\times-0.737)$

$= 0.529 + 0.442 = 0.971$

☁ Overcast (4 days)

D3: Yes

D7: Yes

D12: Yes

D13: Yes

Yes=4 No=0 Total=4

$H = -(4/4)\log_2(4/4) - 0$

$= -(1.0 \times 0) - 0$

$= 0.000$ ✓ Pure!

☂ Rain (5 days)

D4: Yes

D5: Yes

D6: No

D10: Yes

D14: No

Yes=3 No=2 Total=5

$H = -(3/5)\log_2(3/5) - (2/5)\log_2(2/5)$

$= -(0.6\times-0.737) - (0.4\times-1.322)$

$= 0.442 + 0.529 = 0.971$

$$IG(Outlook) = 0.940 - [(5/14)\times0.971 + (4/14)\times0.000 + (5/14)\times0.971] = 0.940 - 0.693 = 0.247$$

Step 4 — IG for Humidity & Wind

HUMIDITY

High (7 days) Yes=3 No=4

$H = -(3/7)\log_2(3/7)-(4/7)\log_2(4/7)$

H(High) = 0.985

Normal (7 days) Yes=6 No=1

$H = -(6/7)\log_2(6/7)-(1/7)\log_2(1/7)$

H(Normal)= 0.592

IG(Humidity) = 0.940 - [(7/14)×0.985 + (7/14)×0.592] = 0.940 - 0.789 = 0.152

WIND

Weak (8 days) Yes=6 No=2

$H = -(6/8)\log_2(6/8)-(2/8)\log_2(2/8)$

H(Weak) = 0.811

Strong (6 days) Yes=3 No=3

$H = -(3/6)\log_2(3/6)-(3/6)\log_2(3/6)$

H(Strong)= 1.000

IG(Wind) = 0.940 - [(8/14)×0.811 + (6/14)×1.000] = 0.940 - 0.892 = 0.048

Step 5 — Comparing All Information Gains

We computed IG for all 4 features. The root node split is chosen by highest IG:

Feature	Subsets	Weighted Entropy	Information Gain	Rank
Outlook	**Sunny/Overcast/Rain**	**0.693**	**0.940 − 0.693 = 0.247**	**1st**
Humidity	High / Normal	0.788	**0.940 − 0.788 = 0.152**	2nd
Wind	Weak / Strong	0.892	**0.940 − 0.892 = 0.048**	4th
Temperature	Hot/Mild/Cool	0.911	**0.940 − 0.911 = 0.029**	3rd

✓ VERDICT — ROOT NODE:

OUTLOOK wins with the highest IG = 0.247 → It becomes the Root of our Decision Tree!

Each branch of Outlook (Sunny / Overcast / Rain) becomes a sub-node — we repeat the process!

Step 6 — Sunny Branch: Which Feature Next?

Sunny subset: D1, D2, D8, D9, D11 → Yes=2, No=3 → H(Sunny) = 0.971 bits

We test remaining features: Temperature, Humidity, Wind

Humidity (Sunny)

High (D1,D2,D8): Yes=0, No=3 → H=0.000

Normal (D9,D11): Yes=2, No=0 → H=0.000

IG = 0.971−[(3/5)×0+(2/5)×0] = 0.971

WINNER — Split on Humidity!

Temperature (Sunny)

Hot (D1,D2): Yes=0, No=2 → H=0.000

Mild (D8): Yes=0, No=1 → H=0.000

Cool (D9,D11): Yes=2, No=0 → H=0.000

IG = 0.971−0 = 0.571

Wind (Sunny)

Weak (D1,D8,D9): Yes=1, No=2 → H=0.918

Strong (D2,D11): Yes=1, No=1 → H=1.000

IG = 0.971−[(3/5)×0.918+(2/5)×1] = 0.019

☀ Sunny → split on HUMIDITY: High → NO ✋ | Normal → YES ✅ (Both pure leaves!)

Step 7 — Rain Branch: Which Feature Next?

Rain subset: D4, D5, D6, D10, D14 → Yes=3, No=2 → H(Rain) = 0.971 bits

We test: Temperature, Humidity, Wind (same idea, but on this smaller subset)

Wind (Rain)

Weak (D4,D5,D10): Yes=3, No=0 → H=0.000

Strong (D6,D14): Yes=0, No=2 → H=0.000

IG = 0.971−[(3/5)×0+(2/5)×0] = 0.971

WINNER — Split on Wind!

Humidity (Rain)

High (D4,D6,D14): Yes=1, No=2 → H=0.918

Normal (D5,D10): Yes=2, No=0 → H=0.000

IG = 0.971−[(3/5)×0.918+0] = 0.420

Temperature (Rain)

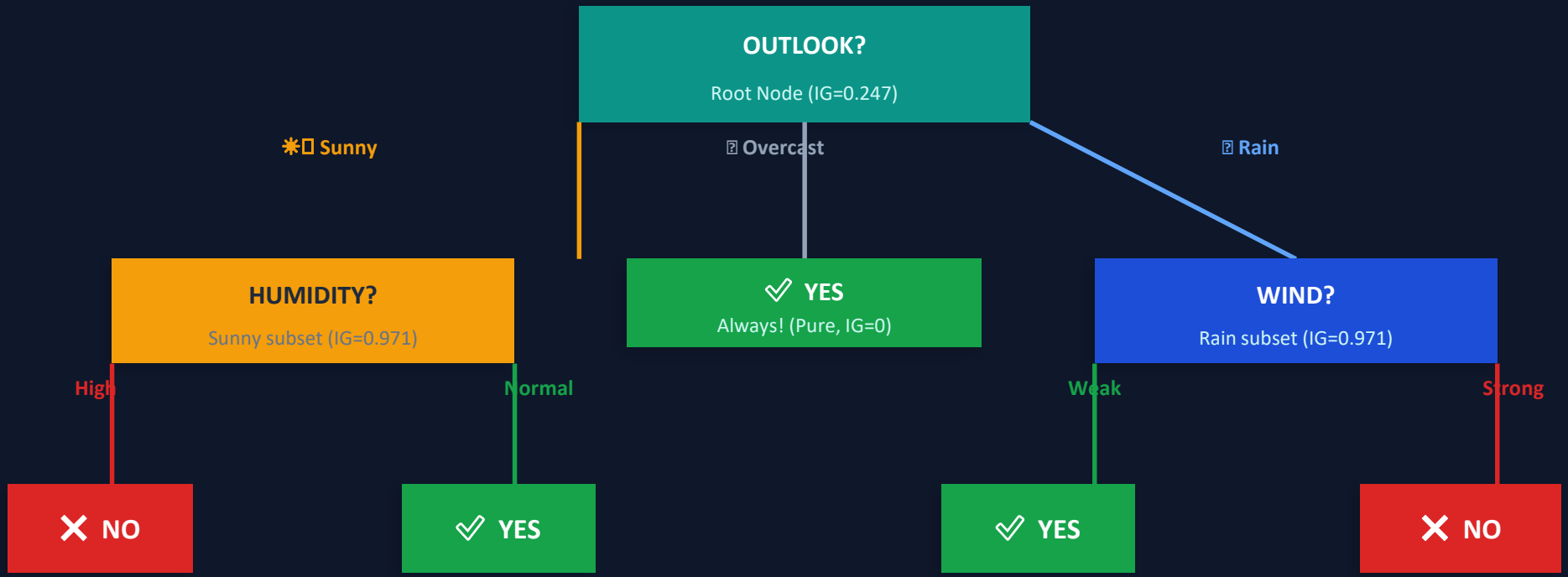
Mild (D4,D10,D14): Yes=2,No=1 → H=0.918

Cool (D5,D6): Yes=1,No=1 → H=1.000

IG = 0.971−[(3/5)×0.918+(2/5)×1] = 0.020

Rain → split on WIND: Weak → YES ✅ | Strong → NO ✋ (Both pure leaves!)

Final Decision Tree



LEGEND:

- Decision Node
- Sub-decision (Sunny)
- Sub-decision (Rain)
- Leaf: YES
- Leaf: NO

Results & Interpretation

Decision Rules Extracted from the Tree:

```
IF Outlook = Overcast  →  YES (Play Tennis)
IF Outlook = Sunny  AND  Humidity = Normal  →  YES
IF Outlook = Sunny  AND  Humidity = High  →  NO
IF Outlook = Rain  AND  Wind = Weak  →  YES
IF Outlook = Rain  AND  Wind = Strong  →  NO
```

Training Accuracy

14 / 14 examples correctly classified on training data!

Overfitting Warning

Tree perfectly fits training data. Use pruning or cross-validation for real generalisation!

Summary & Key Takeaways

01. Entropy measures disorder

H = 0 means pure class, H = 1 means maximum mix. Our root had H = 0.940.

02. Information Gain guides splits

We always pick the attribute with the HIGHEST IG at each node.

03. Root Node = Outlook

IG = 0.247 was the best, so Outlook is the first question the tree asks.

04. Recursive process

After splitting, repeat entropy/IG on each subset until we reach pure leaves.

05. Leaves are final answers

YES/NO decisions — no more splitting needed when a node is pure.

06. Result = 5 clean rules

The tree produces human-readable IF-THEN rules anyone can follow.

Decision Trees are interpretable, powerful, and the foundation of Random Forests & Gradient Boosting!